

Beyond 'Women and Children First': Socioeconomic Status and Survival on the Titanic

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Abstract

The RMS Titanic, one of the most luxurious ocean liners of its time, sank on April 15, 1912, causing the deaths of more than 1,500 people. Using statistical analysis of 1,046 passengers with complete demographic data, this study examines whether survival outcomes align with the stated protocol or reflect the competing influence of socioeconomic class, drawing on historical evidence regarding lifeboat access and ship architecture. Understanding these survival patterns illuminates how institutional crises expose and amplify existing inequalities.

Keywords: Titanic, survival, women and children first, statistics.

Introduction

The RMS Titanic, referred to as the Titanic, was built by the company Harland & Wolff between 1909 and 1912. It had the capacity to accommodate 2,453 passengers and 874 crew members across 10 decks, eight of which were exclusive for passenger use. In the original plans, Alexander Carlisle, the designer of the Titanic, proposed between 32 and 48 lifeboats; this number was later reduced to 20, based on the ship's aesthetics and the regulations of the British Board of Trade safety rules of 1894¹.

The Titanic used the classic three distinct passenger classes commonly seen at the time, which were subsequently subdivided by gender, marital status, nationality, and employment role. The class determined access to specific areas of the liner, and ultimately, to luxury services.

By exploring these two elements of liner structure and class accommodation, we will try to answer the following questions: were the boats enough to provide accommodation for all the passengers on the Titanic, independent of the class in which they traveled? And if the boats were not sufficient, were the guidelines of "women and children first" applied? The third question is whether the chance of survival was correlated to the class to which the passenger belonged. In this last section, we will explore through quantitative analysis, along with historical elements, how it was determined who would gain access to the lifeboats. Each section of the paper will explore one of these questions, concluding with a final summary on the matter.

Lifeboats

One of the most controversial matters on the sinking of the Titanic was the access to lifeboats, which is portrayed as insufficient in popular culture. Did the Titanic violate British regulations? According to the British Board of Trade safety rules of 1894², the appendix to the Life-Saving Appliances Rules based on the

¹ *Formal Investigation into the Loss of the S.S. Titanic: Evidence of Mr. Alexander Carlisle* (Days 19 and 20, 1912)

² British Board of Trade (1912). *Report on the Loss of the "Titanic" (S.S.)*. London: His Majesty's Stationery Office

tonnage of the vessel or in this case the liner. As tonnage increased, the minimum number of required boats also increased; however, the regulations did not account for vessels exceeding 10,000 tons.

Titanic's gross tonnage was approximately 46,000 tons, or 4.6 times the maximum tonnage contemplated by the table. This suggests that the ship's dimensions surpassed the scope of existing safety regulations.

However, the Titanic ended holding 20 lifeboats divided into³:

- 14 Standard Wooden, for capacity of 65 people each;
- 4 Englehardt collapsible boat for capacity of 47 people each;
- 2 emergency cutters for capacity of 40 people each.

Despite the Titanic meeting the requirements of the law, the total capacity of the lifeboats was only 1,178 people. This meant they could accommodate approximately 35% of the ship's maximum capacity of passengers and crew, or 53% of the actual number of people who boarded the vessel for the first and last voyage. This discrepancy left the ship in a critical scenario: in the event of an accident, the lifeboats would be incapable of accommodating the entire population on board. With only 53% capacity, nearly half of those aboard were destined to perish regardless of any selection criterion. This raises the second question of this paper: who would be chosen?

'Women and Children First'

Despite prior historical applications of the 'Women and Children First' protocol, it is commonly accepted that the disaster of the HMS Birkenhead in 1852, where the shortage of lifeboats led officers like Lieutenant-Colonel Alexander Seton to order that women and children be prioritized. This was adopted as a code of conduct, particularly among British sailors known as the Birkenhead drill.

³ Encyclopedia Titanica. (n.d.). Titanic Lifeboat Specifications. Encyclopedia Titanica. <https://www.encyclopedia-titanica.org/lifeboat-specifications.html>

In popular culture, particularly in the 1997 Titanic film, is portrayed that members of the first class, particularly women, were those who boarded the lifeboats, and that some of them were not at full capacity.

Historical records indicate that Captain Edward J. Smith communicated the application of the guideline directly to First Officer William McMaster Murdoch and Second Officer Charles Lightoller, who then transmitted the order through the chain of command⁴. However, the officers' interpretation of that guideline had significant consequences for survival outcomes, as the following sections will show.

Methodological analysis

This study analyzes survival outcomes using data from the Titanic's voyage. The dataset originally contained 1,309 passenger records out of the 1,317 ticketed passengers. However, the analysis was restricted to 1,046 passengers with complete demographic information across all key variables: age, sex, passenger class (1st, 2nd, or 3rd), and survival status.

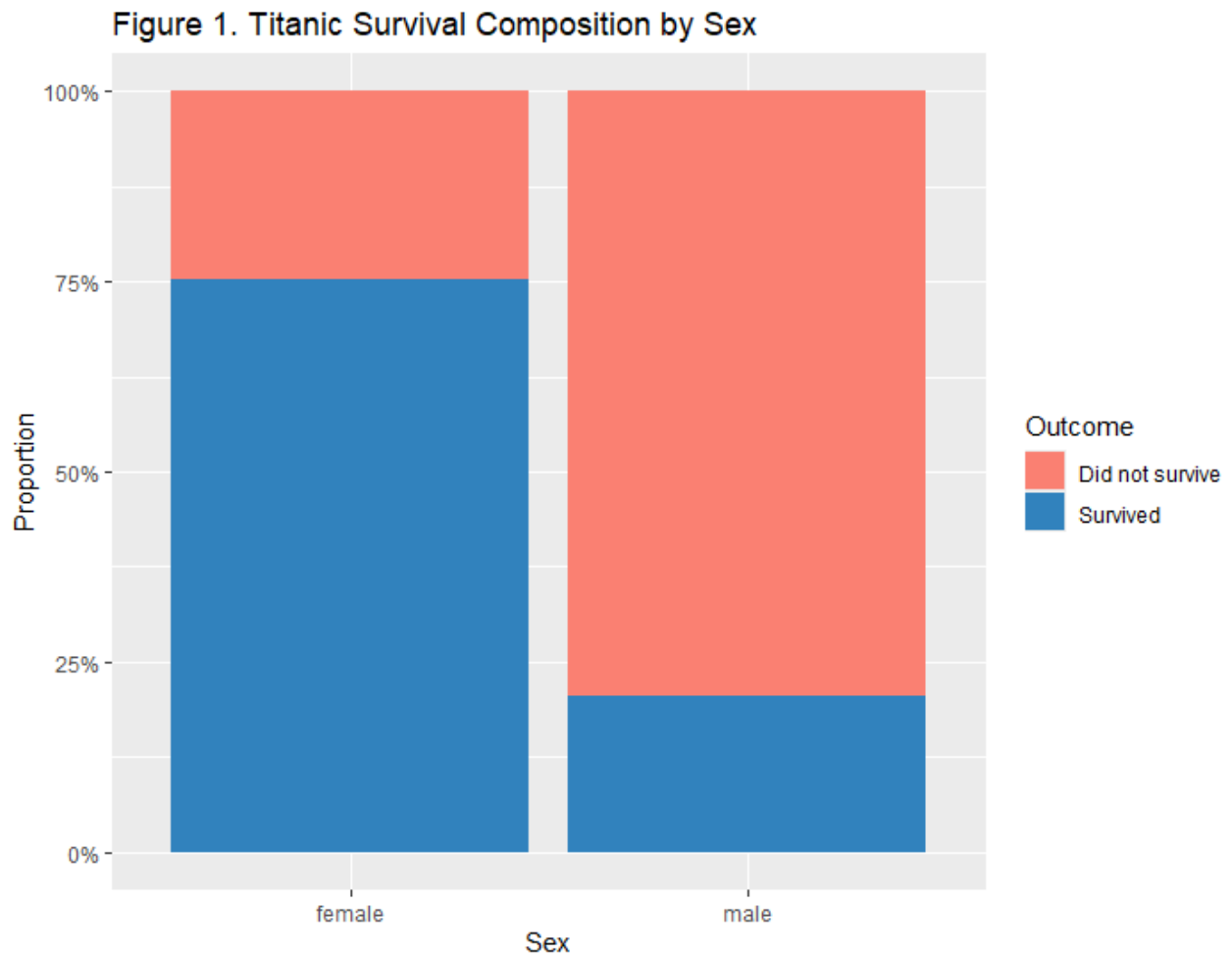
We excluded 263 passengers (20.1% of the original dataset) who lacked age information, as age was essential for testing whether the protocol of "women and children first" was applied across the vessel. This exclusion criterion is justified on two grounds. First, age is a central variable in our hypothesis. Second, the magnitude of missing data (20%) remained within acceptable thresholds for descriptive analysis while preserving statistical stability.

It is important to note that passengers were categorized into two age groups: children (under 12 years) and adults (12 years and older). This threshold reflects the historical context, since in 1912, persons under 12 were typically classified as children in maritime law⁵.

⁴ United States Senate Inquiry (1912). *Titanic Disaster Hearings: Testimony of Charles Lightoller*. Washington: Government Printing Office.

⁵ Encyclopedia Titanica Message Board. "Children On The Titanic." Retrieved from: <https://www.encyclopedia-titanica.org/community/threads/children-on-the-titanic.20096/>

Another important methodological consideration is that this paper does not include crew members in the analysis, since they cannot be categorized into passenger classes, which, once again, is central to our analysis. However, it is important to stress that historical records indicate that approximately 76% of the crew perished. This suggests that the class-based hierarchy of survival extended beyond ticketed passengers.

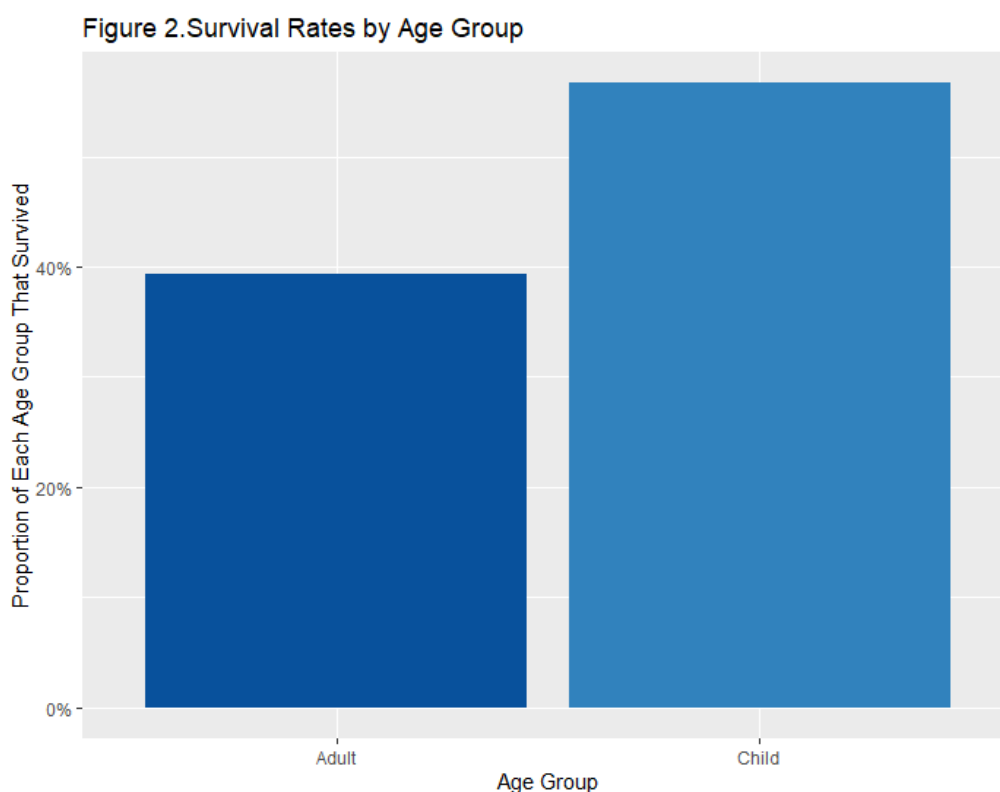


The application of women and children first

Figure 1 shows the survivors divided by gender; we can see the application of the Birkenhead drill as of the 427 passengers of the total that survived, 68% were women. Within the female passenger group itself, 75% survived. If gender had no effect on survival, we would expect the survival rates to be similar across

genders. Instead, the observed difference suggests that gender was associated with the likelihood of survival.

A similar pattern appears in the survival rates of children, suggesting that age was also associated with the likelihood of survival. Figure 2 displays the proportion of each age group that survived, please note that the survival rate computed within each group rather than as a share of total survivors. This distinction matters: because children were far fewer than adults (90 versus 956), expressing children as a fraction of all survivors would understate their relative advantage. Measured within each group, children exceeded adults by 17.3 percentage points (56.6% versus 39.3%).



However, among children, survival rates do not differ substantially by sex: girls survived at 59.2% and boys at 54.17%, a gap of roughly seven percentage points. This near parity stands in sharp contrast to the adult population, where adult women survived at 77% and adult men at only 17.8% of their own

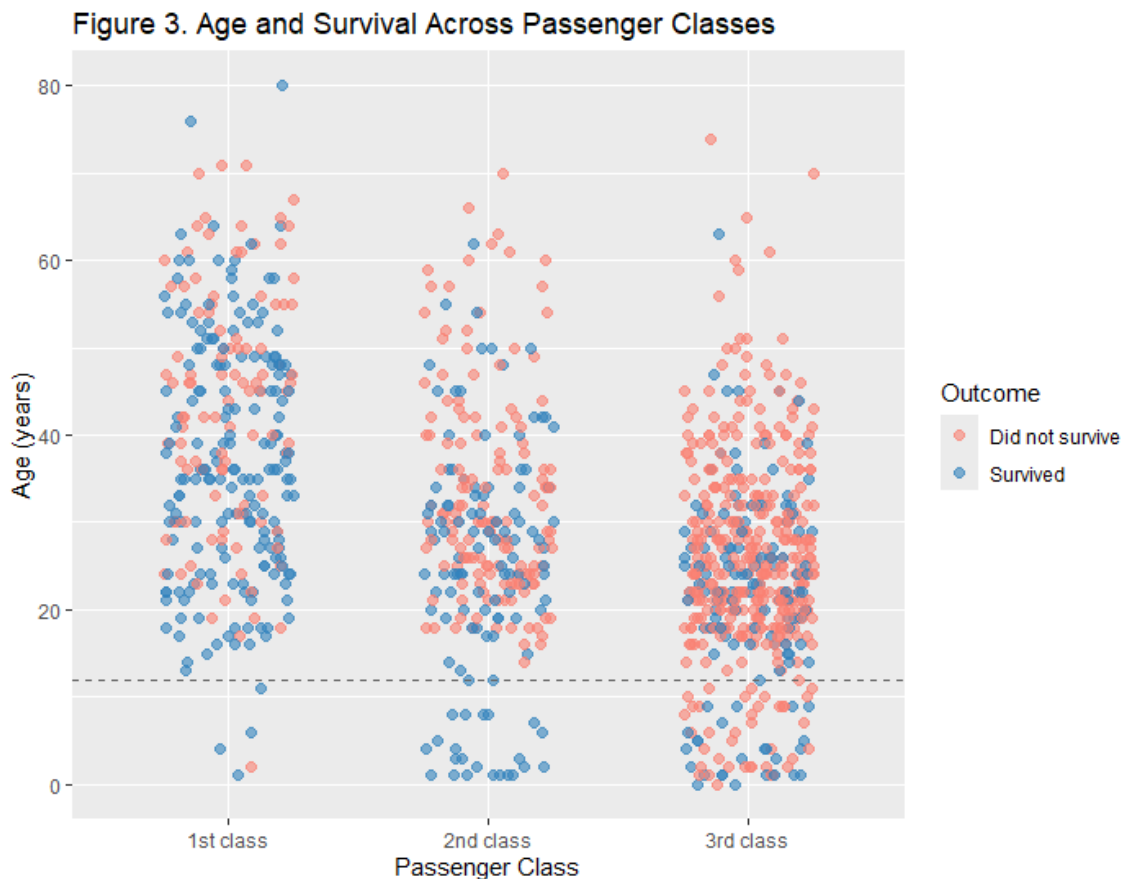
group. The relatively equal outcomes for boys and girls suggest that, within childhood, sex was not a decisive factor in survival. The decisive gender divide emerges only among adults.

Taken together, these findings show that both gender and age significantly shaped survival outcomes. Yet they also point to a further question: was the Birkenhead drill applied consistently across all passenger classes? The following section explores how socioeconomic status intersected with age and gender to shape survival outcomes.

Socioeconomic Status: the hidden factor

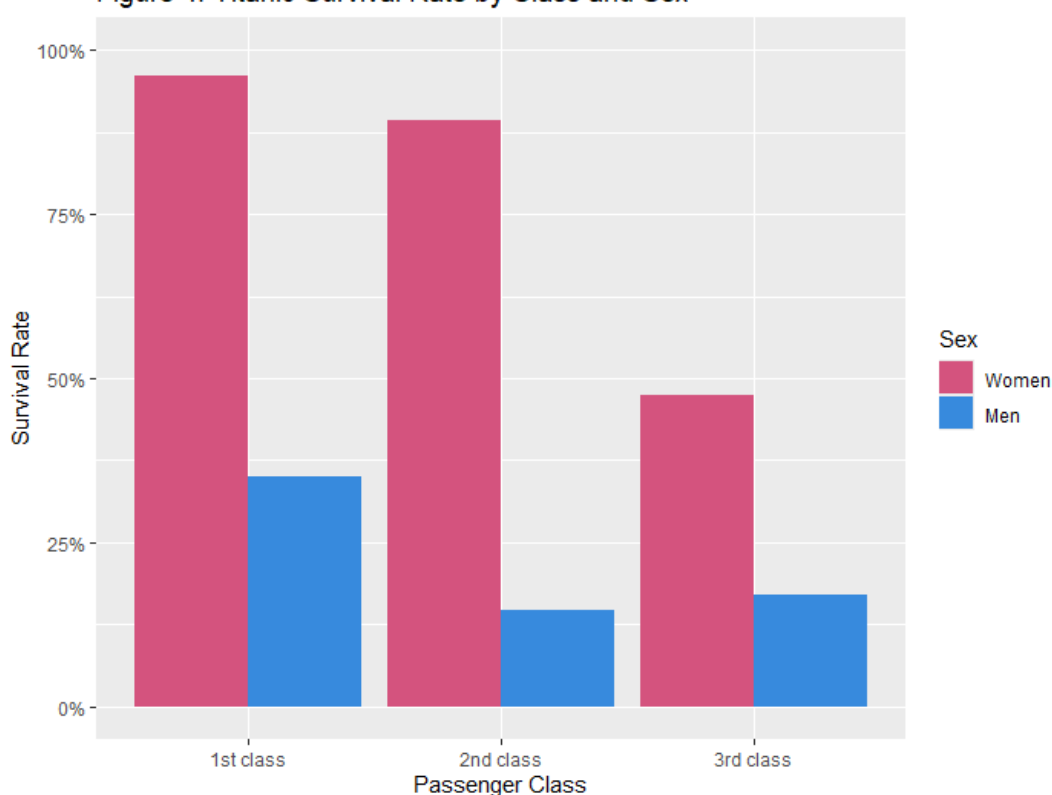
As shown in the previous section, both gender and age played an important role in determining access to the limited number of lifeboats. However, the drill did not specify the order in which the women, particularly those who were not caregivers, and the men who survived should board the lifeboats. Furthermore, the drill did not specify that only women and children were the ones to board the lifeboats, rather was a guideline to execute priority. As stated before, the lifeboats were only capable of holding 53% of the people that traveled on the first Titanic voyage.

After considering other variables that might explain access to the lifeboats, passenger class emerges as a key explanatory factor. As shown in Figure 3, survival varied markedly across passenger classes, with the red dots representing passengers who did not survive. The same figure also includes the 12-year threshold that, as explained earlier, defines those classified as children. Although the methodology excludes 20% of the original dataset because of missing age data, Figure 3 reflects a historic fact: among first- and second-class passengers, only one child did not survive. Thus, while survival among children does not differ substantially by sex, it does vary by class.



Regarding gender, as shown in Figure 4, there is a substantial difference in survival rates among women by class, with third-class women surviving at much lower rates than women in the upper classes. A class gradient also appears among men: although survival rates for second, and third-class men are both low, first-class men show a markedly higher survival rate than either group. These patterns suggest that class shaped access to lifeboats in ways that went beyond the formal priority given to women and children.

Figure 4. Titanic Survival Rate by Class and Sex



One explanation for the higher survival rate among first-class passengers is their proximity to the upper decks, especially the Boat Deck or Top Deck, where all the lifeboats were located. This area was primarily accessible to first-class and second-class passengers. However, both classes were segregated and did not allow them to mix in normal circumstances. It is important to stress that the different classes did not share common spaces other than the Sunday Church Service, and some spaces during the boarding process⁶.

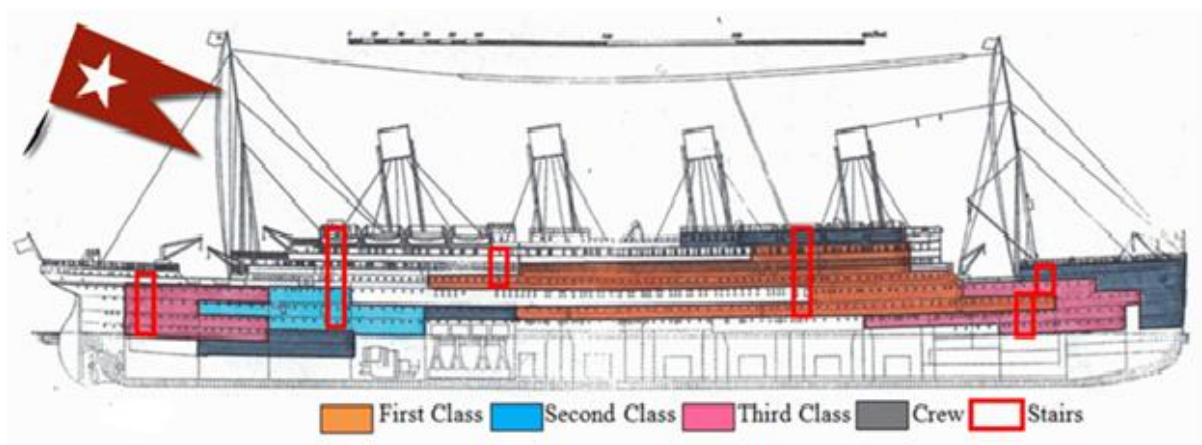
Historical records suggest that second-class passengers faced difficulties accessing the lifeboats due to Second Officer Charles Lightoller's interpretation of the Birkenhead drill⁷. Lightoller, who was

⁶ Just first and second shared common areas. Due to the In the Edwardian era, strict immigration and sanitary laws, such as the United States Steerage Act of 1819 and subsequent public health regulations, mandated the physical segregation of third-class passengers from upper classes. The primary objective of "not mixing" was to prevent the spread of infectious diseases, such as trachoma, lice, and typhus, that were prevalent among incoming immigrants.

⁷ Bowdoin College, History 2203. "Disproportionate Devastation." Fall 2020. Retrieved from: <https://courses.bowdoin.edu/history-2203-fall-2020-kmoyniha/reflection/> Citing: Levinson, 2012, p. 149.

responsible for loading the portside lifeboats, allowed only women and children to board, and several of those boats were launched below full capacity. This could explain the lower survival rates of males, particularly those in second class.

Figure 5. Titanic's class distribution



Regarding the survival of children of the third class, as shown in Figure 5, due to health considerations, the first and second classes did not share common spaces with the third class, as it was strictly forbidden. Furthermore, the third class did not have direct vertical access to the boat deck. They had to navigate a complex series of stairwells, fire doors, and corridors. The third class was composed partially of immigrants of different nationalities, some of them did not speak English which made it even harder to understand signaling to access the emergency routes.

Third-class passengers were also segregated by gender. Single men were housed in the extreme forward section (the bow), whereas single women and families were in the extreme aft section (the stern). Although the stern flooded later than the bow, navigating the ship's complex passageways with children may help explain why this group had lower survival rates than others, especially because the access to the boat deck required the access to the Aft Well Deck on C-Deck, and from there each family had to find their way to the boat deck.

While women and children showed higher survival probabilities than adult men, this advantage becomes conditional on socioeconomic position once passenger class is considered. The following table illustrates how survival probabilities shift within each demographic group across classes.

Table 1. Survival probabilities by demographic group and passenger class.

Women (overall)	75.26%
Women, 1st class	96.97%
Women, 2nd class	88.04%
Women, 3rd class	47.54%
Men (overall)	20.52%
Men, 1st class	33.33%
Men, 2nd class	8.16%
Men, 3rd class	15.19%
Girls (Overall)⁸	59.52%
Girls, 1st class ⁹	NA
Girls, 2nd class	100.00%
Girls, 3rd class	46.67%
Boys (Overall)¹⁰	54.17%
Boys, 1st class	100.00%
Boys, 2nd class	100.00%
Boys, 3rd class	33.33%

⁸ Please notice that due the size of the sample this is a weighted rate.

⁹ Girls, 1st class excluded (n=1, statistically unreliable).

¹⁰ Please notice that due the size of the sample this is a weighted rate.

This table synthesizes the main patterns discussed in the previous sections. Age and gender were important factors in survival, but passenger class also emerges as a decisive variable across demographic groups. One exception concerns first-class girls, who are not included in the rate analysis because that category contains only a single case. In that instance, the child's family intentionally chose not to board a lifeboat despite having the opportunity to do so, reportedly because a relative was missing.

To assess more formally whether survival was independent of passenger class, this paper applies the multiplication rule of probability. If survival and class were independent, the joint probability $P(\text{survived} \cap \text{first class})$ would equal $P(\text{survived}) \times P(\text{first class})$. In this dataset, $P(\text{survived}) = 40.8\%$ and $P(\text{first class}) = 27.2\%$, which yields an expected joint probability of 11.1% under independence. The observed joint probability, however, is 17.3%, a difference that indicates survival and passenger class were not independent. A similar pattern appears for sex: under independence, $P(\text{survived} \cap \text{female})$ would be 16.0%, whereas the observed value is 27.9%.

Taken together, these probability comparisons reinforce the central claim of this paper: survival aboard the Titanic cannot be explained by a single rule or demographic factor alone. Instead, the evidence points to an interaction between formal social norms and the stratified structure of the ship itself, making class an essential dimension of any broader interpretation of the disaster.

Conclusion

The sinking of the Titanic was a complex historical event shaped by multiple factors, and any analysis of survival must take that complexity into account. This paper has shown, through a focused statistical analysis grounded in a historical dataset, that survival was not determined solely by the principle of "women and children first." Rather, passenger class, which functioned as a proxy for socioeconomic status, also played a decisive role in shaping access to the lifeboats and, ultimately, survival outcomes.

The evidence presented throughout this paper suggests that the Titanic disaster did not simply reveal whether the Birkenhead drill was followed, it also exposed how deeply social hierarchy structured

the conditions of survival. The shortage of lifeboats made some form of selection unavoidable, but that selection did not occur in a socially neutral environment. Access to upper decks, proximity to lifeboats, the physical segregation of third-class passengers, and the differing interpretations of evacuation priority all interacted to produce unequal outcomes. In that sense, class did not replace the rule of “women and children first,” but rather conditioned how fully that rule could operate in practice. The result was that women and children in the upper classes benefited most from the norm, while many third-class passengers, especially families navigating complex routes from isolated sections of the ship, faced structural disadvantages long before they reached the boat deck. Ultimately, the Titanic can be understood not only as a maritime tragedy, but also as a moment in which the social order of the Edwardian world remained visible even in disaster.

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